

Name.....

Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- NOVEMBER 2022

Fifth Semester – (Common to CSE and AD)

CST 303 COMPUTER NETWORKS

Time: 2 hours

Max. Marks: 50

PART A

Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
✓1	Define Network Software. List the design issues of Layers.	3	K1	CO1 & K2
✓2	Write a short note on Network Topologies.	3	K1	CO1 & K2
3/✓	Consider a code with only four valid code words: 0000000000, 0000011111, 1111100000, and 1111111111. If the code word arrived is 0000000111 then what will be the original code word transmitted?	3	K3	CO2 & K3
④	Draw the Frame format of Ethernet. Ethernet frames must be at least 64 bytes long to ensure that the transmitter is still going in the event of a collision at the far end of the cable. Fast Ethernet has the same 64-bytes minimum frame size but can get the bits out ten times faster. How is it possible to maintain the same minimum frame size?	3	K3	CO4 & K2
⑤	Define Routing Algorithms. What are the 2 general categories of Routing Algorithms?	3	K1	CO5 & K3

PART B

Answer any 2 questions (7 x 2 = 14 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
6/✓	Differentiate ISO/OSI and TCP/IP Protocol Suites	7	K2	CO1 & K2
7	a) Explain in detail about the different Transmission Media	5	K1	CO2 & K3
	b) If there are n devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively.	2	K3	CO2 & K3
✓	Create Digital Signals from the following digital data using Manchester and Differential Manchester Encodings. a. 00110101001 b. 11011011011 c. 10001010001	7	K2	CO2 & K3

PART C

Answer any 2 questions (7 x 2 = 14 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
9	A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . a. Calculate the CRC value. b. What is the actual bit string transmitted? c. If the received data is 110101101 <u>0</u> 1100, check whether the received data is correct or not?	7	K3	CO3 & K2
10	With the help of diagrams, explain CSMA/CD and MACA Protocols.	7	K1	CO4 & K2
11	a) Explain the different Framing Techniques used in Data Link layer. b) If Flag byte = 01111110 and ESC byte = 11101110, show how byte stuffing can be done to the following data. 110000001111011100111111011101111	4	K1	CO3 & K2
		3	K3	CO3 & K2

PART D

Answer all questions (7 x 1 = 7 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
12	a) Compare Packet Switching and Circuit Switching b) Consider the given network uses Packet Switching for transmission of 5 packets between Node A and Node C and the time taken for transmission between 2 directly connected nodes is 2ms. If the transmission starts at time t_0 and at time t_5 the link between Node A and Node D gets down, which will be the path taken by each of the packets to reach the destination?	3	K2	CO5 & K3
		4	K3	CO5 & K3

$A \rightarrow B$
 $A \rightarrow D$
 $B \rightarrow C$
 $D \rightarrow E$

$A \rightarrow B$
 $D \rightarrow B$
 $A \rightarrow D$
 $A \rightarrow C$
 B

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ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE SECOND INTERNAL TEST- DECEMBER 2022

Fifth Semester – (Common to CSE and AD)

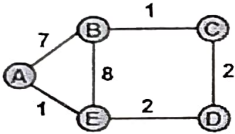
CST 303 COMPUTER NETWORKS

Time: 2 hours

Max. Marks: 50

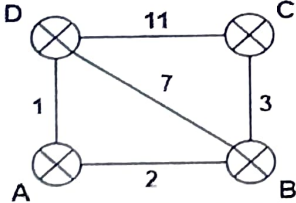
PART A

Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1/	Using Dijkstra's Shortest path algorithm, find the shortest distances and path to all the nodes from Source A. 	3	K3	CO6 & K2
2	Explain any 3 Closed Loop Congestion control mechanisms	3	K1	CO5 & K3
3	Explain the structure and classes of IPV4. Write the Class and Default mask of the following IP Addresses 1. 192.168.178.1 2. 102.168.212.226	3	K2	CO6 & K2
4/	State the Optimality Principle.	3	K1	CO5 & K3
5	Draw the UDP header format and explain the fields.	3	K1	CO6 & K2

PART B

Answer any 2 questions (7 x 2 = 14 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
6	Describe the two Traffic Shaping methods for improving Quality of Service. Specify how these methods can be used so as to improve QoS.	7	K2	CO5 & K3
7/	a) Explain Distance Vector Routing Algorithm b) The following network uses Distance Vector Routing . Create Routing tables for each node. Explain the steps. 	2	K1	CO5 & K3
		5	K3	CO5 & K3

8	Explain how Routing table is created by node D using Link State Routing Algorithm.	7	K3	CO5 & K3
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PART C

Answer any 2 questions (7 x 2 = 14 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
9	a) If the following is an ICMP header, give the value of ICMP checksum explain the steps in checksum calculation.	4	K3	CO6 & K2
	b) Explain the ICMP Error Reporting Messages.	3	K1	
10	With the help of Packet formats, explain how ARP and RARP protocols are used for Address Mapping.	7	K1	CO6 & K2
11	a) If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?	2	K3	CO6 & K2
	b) Suppose a network with IP Address 192.16.20.0. is divided into 4 subnets, find number of hosts per subnet. Also for each of the subnets, find- - Subnet Address - First Host ID - Last Host ID - Broadcast Address	5	K3	CO6 & K2

PART D

Mandatory (7 x 1 = 7 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
12	a) Draw the TCP header format and explain its fields.	2	K1	CO6 & K2
	b) Elaborate on phases in TCP Connection with the help of appropriate diagrams	5	K2	

Reg No: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST 303**Course Name: COMPUTER NETWORKS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

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|----|--|---|
| 1 | Illustrate layering principle with necessary diagrams. | 3 |
| 2 | What is the propagation delay if distance between two points is 12,000 km?
Assume propagation speed to be 2.4×10^8 m/s in cable. | 3 |
| 3 | Show how the data field is managed in the Ethernet frame format. | 3 |
| 4 | Differentiate between bridges and switches. | 3 |
| 5 | Illustrate the optimality principle in routing. | 3 |
| 6 | Identify two methods by which multicast routing is handled by network layer. | 3 |
| 7 | In IP, the checksum covers only the header and not the data. Identify the reason for selecting this design. | 3 |
| 8 | Differentiate between Open loop and Closed loop approaches for congestion control. Give one example for each. | 3 |
| 9 | Why is Transport layer called true End to End layer? Why is flow control and error control used in Transport layer in addition to data link layer? | 3 |
| 10 | Describe the ports used by the FTP with suitable diagram. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|---|
| 11 | a) Justify why TCP/ IP reference model is called as a protocol stack model.
Describe TCP/IP model with suitable diagrams. | 8 |
| | b) Compare LAN, MAN, WAN. | 6 |
| 12 | a) Compare the types of networks formed based on topology. Discuss the advantages and disadvantages of each type. | 8 |
| | b) Illustrate the construction of optical fiber and justify how this reduces the interference. Draw the structure of optical fiber. | 6 |

Module -2

- 13 a) Summarize about the high-speed LANs specified in IEEE standard. 6
 b) Compare Go back N protocol with Selective Repeat protocol using required diagrams. 8
- 14 a) Discuss about the frame formats of HDLC protocol. 8
 b) Write short notes on any two carrier sense multiple access protocols used in IEEE standards. 6

Module -3

- 15 a) Compare the features of link state routing with distance vector routing. 8
 b) Differentiate between the implementation of datagram subnet and virtual circuit subnet. 6
- 16 a) What is count to infinity problem? Describe two techniques to solve the count to infinity problem in distance vector routing algorithm. 6
 b) Identify and describe the scheduling techniques to improve the Quality of Services (QoS). 8

Module -4

- 17 a) Illustrate the sub-netting concept. A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Find the number of subnets possible and hosts which can be connected in each subnet. 8
 b) Identify the characteristics of BGP. 6
- 18 a) Describe how does OSPF perform routing in larger networks? Also explain the different types of OSPF messages. 7
 b) Explain the purposes of using ARP and RARP in the network layer. Also describe the working of each. 7

Module -5

- 19 a) Describe the TCP congestion control approaches with necessary diagrams. 8
 b) Demonstrate the UDP segment structure. 6
- 20 a) Summarize the architecture of electronic mail system with neat diagram. 9
 b) Identify and describe the DNS attacks. 5

